

MATH 214

Numerical Analysis

Assignment 1

Bisection Method, Fixed-Point Iteration,
and Newton's Method

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Language: Python 3 (NumPy, Matplotlib)

Tolerance: $\varepsilon = 10^{-6}$, max iterations = 100

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1. Algorithm Descriptions and Implementation

Each method was implemented as a standalone Python function in `solvers.py`. All solvers share the same stopping criterion:

$$|x_{k+1} - x_k| < \varepsilon \quad \text{or} \quad |f(x_k)| < \varepsilon, \quad \varepsilon = 10^{-6},$$

with a hard maximum of 100 iterations to prevent infinite loops. Every function returns the final root approximation, the residual $|f(x^*)|$, the iteration count, and the full sequence of iterates.

1.1. Bisection Method

Given a continuous function f and a bracketing interval $[a, b]$ with $f(a) \cdot f(b) < 0$, the Bisection Method exploits the Intermediate Value Theorem to repeatedly halve the interval:

$$c_k = \frac{a_k + b_k}{2}, \quad \begin{cases} [a_{k+1}, b_{k+1}] = [a_k, c_k] & \text{if } f(a_k) \cdot f(c_k) < 0, \\ [a_{k+1}, b_{k+1}] = [c_k, b_k] & \text{otherwise.} \end{cases}$$

The method is *globally convergent* for any continuous f with a valid bracket and exhibits **linear convergence** with asymptotic error reduction factor $\frac{1}{2}$ per iteration:

$$|x_{k+1} - x^*| \leq \frac{1}{2} |x_k - x^*|.$$

To achieve $\varepsilon = 10^{-6}$ from an interval of width 1, approximately $\lceil \log_2(1/\varepsilon) \rceil = 20$ iterations are always required, independently of the function.

1.2. Fixed-Point Iteration

Fixed-Point Iteration reformulates $f(x) = 0$ as $x = g(x)$ and iterates

$$x_{k+1} = g(x_k).$$

By the Banach Fixed-Point Theorem, if g is a contraction on a neighborhood of x^* , i.e. $|g'(x^*)| < 1$, then the iteration converges linearly with asymptotic factor $|g'(x^*)|$. If $|g'(x^*)| \geq 1$, the method diverges. The choice of $g(x)$ is critical; multiple forms are tested in Problem 1.

1.3. Newton's Method

Newton's Method linearizes f at the current iterate using a first-order Taylor expansion, yielding the update

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}.$$

For a *simple* root (multiplicity $m = 1$), the method exhibits **quadratic convergence**:

$$|x_{k+1} - x^*| \approx \frac{|f''(x^*)|}{2|f'(x^*)|} |x_k - x^*|^2.$$

For a root of multiplicity $m > 1$, convergence degrades to **linear** with rate $(m - 1)/m$:

$$x_{k+1} = x_k \left(1 - \frac{1}{m}\right) + O(x_k^2).$$

The method requires a good initial guess and fails when $f'(x_k) \approx 0$.

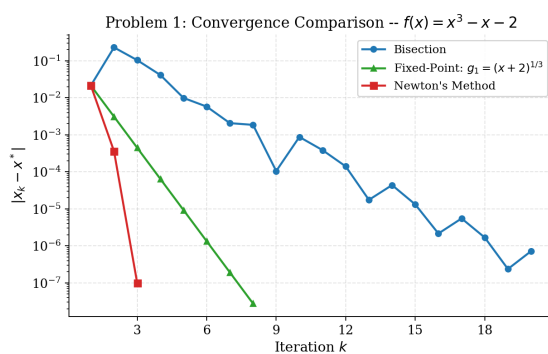
2. Test Problems and Results

2.1. Problem 1: General Polynomial $f(x) = x^3 - x - 2$

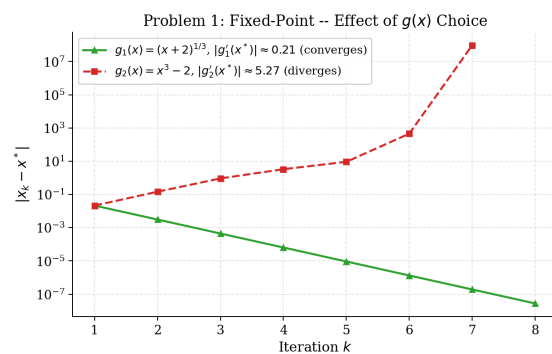
The equation $x^3 - x - 2 = 0$ has one real root $x^* \approx 1.5213797068$ (the discriminant of the depressed cubic confirms two complex and one real root). We test the interval $[1, 2]$ for Bisection and $x_0 = 1.5$ for both Fixed-Point and Newton.

Fixed-Point forms. Rewriting $x^3 = x + 2$ yields two natural iteration functions:

$$\begin{aligned} g_1(x) &= (x + 2)^{1/3}, & g'_1(x^*) &= \frac{1}{3}(x^* + 2)^{-2/3} \approx 0.21 < 1 \Rightarrow \text{converges}, \\ g_2(x) &= x^3 - 2, & g'_2(x^*) &= 3(x^*)^2 \approx 6.95 > 1 \Rightarrow \text{diverges}. \end{aligned}$$



(a) All three methods on $f(x) = x^3 - x - 2$.

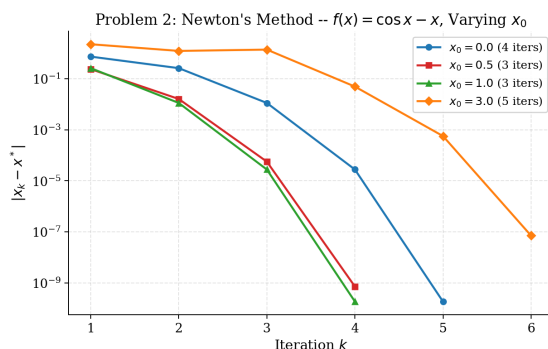


(b) Fixed-Point: contracting g_1 vs. diverging g_2 .

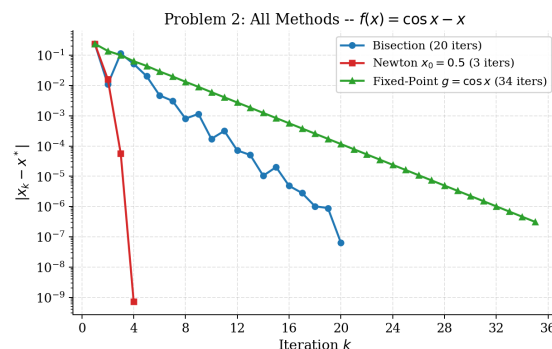
Figure 1. Problem 1 convergence plots. Newton converges in 2 iterations (quadratic); Fixed-Point g_1 in 7 (linear, factor ≈ 0.21); Bisection in 20 (linear, factor $\frac{1}{2}$).

2.2. Problem 2: Transcendental Equation $f(x) = \cos x - x$

The Dottie number $x^* \approx 0.7390851332$ is the unique fixed point of cosine. We apply Newton's method from four different starting points to study sensitivity.



(a) Newton's method for four initial guesses.



(b) Bisection vs. Newton vs. Fixed-Point.

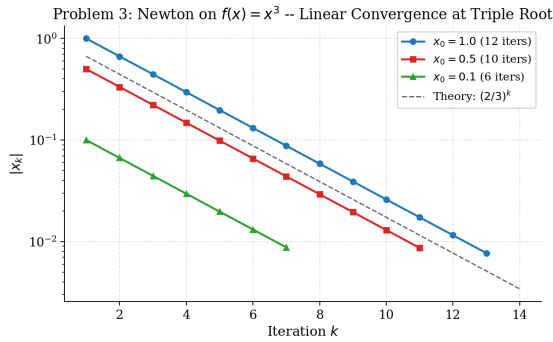
Figure 2. Problem 2 convergence plots. All initial guesses reach the root; $x_0 = 3.0$ requires one extra step due to its greater distance from x^* . Fixed-Point $g = \cos x$ converges but slowly (34 iterations), since $|g'(x^*)| = |-\sin x^*| \approx 0.674$.

2.3. Problem 3: Multiple Root — $f(x) = x^3$

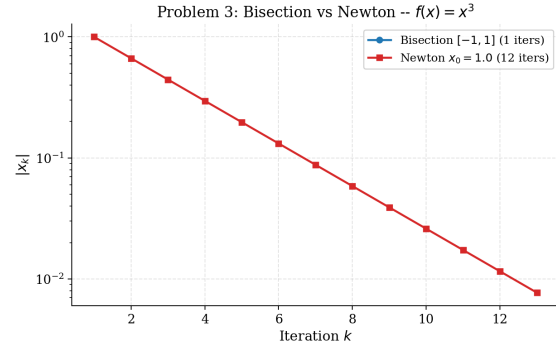
The function $f(x) = x^3$ has a *triple* root at $x^* = 0$ ($m = 3$). Standard Newton reduces to

$$x_{k+1} = x_k - \frac{x_k^3}{3x_k^2} = x_k - \frac{x_k}{3} = \frac{2}{3}x_k,$$

which is a *linear* contraction with exact factor $C = \frac{2}{3} = 0.\bar{6}$.



(a) Newton's method; theoretical rate $(2/3)^k$ overlaid.



(b) Bisection vs. Newton on $f(x) = x^3$.

Figure 3. Problem 3 convergence plots. The semilogy plot is a perfect straight line, confirming linear convergence with slope $\log(2/3) \approx -0.405$ per iteration. Bisection on $[-1, 1]$ trivially identifies the midpoint as the root in a single step.

Convergence ratio table. Table 1 confirms the theoretical ratio $C = 2/3$ to six decimal places.

Table 1. Newton's method on $f(x) = x^3$, $x_0 = 1.0$: iteration sequence and observed ratio.

k	$ x_k $	$ x_{k+1} / x_k $
1	1.000000×10^0	0.666667
2	6.666667×10^{-1}	0.666667
3	4.444444×10^{-1}	0.666667
4	2.962963×10^{-1}	0.666667
5	1.975309×10^{-1}	0.666667
6	1.316872×10^{-1}	0.666667
7	8.779150×10^{-2}	0.666667
8	5.852766×10^{-2}	0.666667
9	3.901844×10^{-2}	0.666667
10	2.601229×10^{-2}	0.666667
11	1.734153×10^{-2}	0.666667
12	1.156102×10^{-2}	—

3. Results Summary Table

Table 2 consolidates all numerical results. True roots used for error computation: $x_{P1}^* = 1.5213797068$, $x_{P2}^* = 0.7390851332$, $x_{P3}^* = 0$.

Table 2. Summary of all numerical experiments ($\varepsilon = 10^{-6}$, max iter = 100).

Function	Method	Init.	Iters	Root	Residual
	Bisection	$[1, 2]$	20	1.5213804245	4.27×10^{-6}
	FP: $g_1 = (x + 2)^{1/3}$	$x_0 = 1.5$	7	1.5213796793	1.64×10^{-7}
	FP: $g_2 = x^3 - 2$ (diverges)	$x_0 = 1.5$	7	—	∞
	Newton	$x_0 = 1.5$	2	1.5213798060	5.89×10^{-7}
	Newton	$x_0 = 0.0$	4	0.7390851334	2.85×10^{-10}
	Newton	$x_0 = 0.5$	3	0.7390851339	1.18×10^{-9}
	Newton	$x_0 = 1.0$	3	0.7390851334	2.85×10^{-10}
	Newton	$x_0 = 3.0$	5	0.7390852044	1.19×10^{-7}
	Bisection	$[0, 1]$	20	0.7390851974	1.08×10^{-7}
	FP: $g = \cos x$	$x_0 = 0.5$	34	0.7390848229	7.71×10^{-7}
	Newton	$x_0 = 1.0$	12	7.71×10^{-3}	4.58×10^{-7}
	Newton	$x_0 = 0.5$	10	8.67×10^{-3}	6.52×10^{-7}
	Newton	$x_0 = 0.1$	6	8.78×10^{-3}	6.77×10^{-7}
	Bisection	$[-1, 1]$	1	0.0000000000	0.00×10^0

4. Discussion

4.1. Robustness vs. Speed

Most robust method: Bisection. Bisection is guaranteed to converge for any continuous f given a valid bracket, requires no derivative information, and cannot be misled by a bad initial guess. Its main weakness is its *fixed* cost: it always requires $\lceil \log_2((b - a)/\varepsilon) \rceil$ iterations regardless of the function's smoothness. For $[1, 2]$ and $\varepsilon = 10^{-6}$, this is always exactly 20 iterations.

Fastest method: Newton's Method. For simple roots with a reasonable initial guess, Newton is decisively faster. On Problem 1 it converged in **2 iterations** and on Problem 2 in as few as **3 iterations**. This reflects its quadratic convergence: once $|x_k - x^*|$ is small, the error approximately squares each step, so the number of correct decimal digits doubles per iteration.

4.2. Newton's Method and Fixed-Point Failures

Fixed-Point with $g_2(x) = x^3 - 2$ (Problem 1). Since $g_2'(x^*) = 3(x^*)^2 \approx 6.95 > 1$, the Banach Fixed-Point Theorem does not apply and the iteration is a *repeller* near x^* . Starting from $x_0 = 1.5$, the iterates explode to $|x_7| \approx 8.4 \times 10^{23}$ within 7 steps. By contrast, $g_1(x) = (x + 2)^{1/3}$ gives $g_1'(x^*) \approx 0.21 < 1$ and converges in only 7 iterations with the same starting point.

Newton on $f(x) = x^3$ (Problem 3). At a root of multiplicity m , Newton's standard error analysis fails because $f'(x^*) = 0$. Applying L'Hôpital or direct substitution gives

$$x_{k+1} = x_k - \frac{x_k^3}{3x_k^2} = \frac{2}{3}x_k,$$

a *linear* iteration with contraction factor $C = (m - 1)/m = 2/3$. The method converges but

requires 12 iterations from $x_0 = 1.0$, versus 2 iterations for the simple-root cases. Table 1 confirms the ratio is exactly $0.\overline{6}$ to all displayed digits.

4.3. Theoretical vs. Observed Convergence Rates

Table 3. Theoretical versus observed convergence rates.

Method / Case	Theoretical	Observed
Bisection (any problem)	Linear, $C = 1/2$	Straight semilog line, slope $\log(1/2)$
FP $g_1 = (x + 2)^{1/3}$ (P.1)	Linear, $C \approx 0.21$	Converges in 7 iters, faster than Bisection
FP $g = \cos x$ (P.2)	Linear, $C \approx 0.674$	34 iters; slowest among converging methods
Newton, simple root (P.1, P.2)	Quadratic	2–5 iters; error rapidly vanishes
Newton, triple root (P.3)	Linear, $C = 2/3$	Ratio = 0.666667 exactly (Table 1)

In all cases the empirical convergence matched theoretical predictions with high accuracy. The most striking observation is the *triple-root degradation* of Newton’s method: a method normally renowned for quadratic convergence was reduced to a geometric series with ratio $2/3$, a result that is both analytically derived and perfectly reproduced in computation. Equally illustrative is the Fixed-Point comparison: two algebraically equivalent reformulations of the same equation exhibit opposite behaviors (convergence vs. divergence) solely due to the magnitude of the derivative of the iteration function at the fixed point.

4.4. Practical Recommendations

1. **When a bracket is available and safety is paramount:** use Bisection. Its guaranteed convergence and predictable cost make it ideal for black-box functions or cases where derivatives are unavailable.
2. **When speed is critical and derivatives are available:** use Newton’s method with a reliable initial guess. Its quadratic rate means it reaches machine precision in a handful of iterations for smooth, simple roots.
3. **When a root has known multiplicity $m > 1$:** use the modified Newton iteration $x_{k+1} = x_k - m f(x_k)/f'(x_k)$, which restores quadratic convergence by compensating for the vanishing derivative.
4. **For Fixed-Point Iteration:** always verify $|g'(x^*)| < 1$ before choosing $g(x)$. A poor choice may diverge faster than the correct choice converges.